

Equine Lameness Location Prediction

Using Inertial Sensor Data and Clinical Medical Records

A Multi-Phase Machine Learning Study

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May 2026

92.6%

Final Accuracy

100%

Nonlimb Recall

99%

Proximal Recall

209

Total Features

Abstract

Equine lameness is a leading cause of reduced performance and welfare concerns in sport and working horses. Accurate localization of the lame limb region — distal (fetlock/hoof), middle (carpus/tarsus), proximal (shoulder/hip) or non-limb (back/spine) — is critical for guiding targeted diagnostic workup. This report presents a comprehensive machine learning study that combines 16-channel inertial sensor gait data with structured clinical records and free-text veterinary notes to predict lameness location across 514 horses (10,568 stride observations). Through systematic experimentation across eight model iterations — from a baseline LightGBM (49.3%) to a fully optimized two-stage stacking ensemble (92.6%) — we demonstrate how architectural choices, feature engineering, and hyperparameter optimization each contribute distinct performance gains. The final model achieves 100% recall on the clinically important non-limb class and 99% recall on proximal cases, reducing the risk of missed diagnoses.

1. Clinical Motivation & Background

1.1 Clinical Context

Lameness is among the most prevalent conditions in equine medicine, affecting an estimated 60–70% of horses at some point in their working life. Accurate identification of the anatomical region responsible for lameness is the first — and often most challenging — step in the diagnostic pathway. Physical examination, including flexion tests and nerve blocks, can be time-consuming, expensive, and invasive. Objective gait analysis using body-mounted inertial measurement units (IMUs) has emerged as a reliable, non-invasive complement to subjective lameness evaluation. However, translating raw sensor asymmetry scores into a specific anatomical location remains an open challenge.

1.2 Study Objectives

- Develop a machine learning classifier capable of predicting lameness location (distal, middle, proximal, non-limb) from IMU-derived gait parameters and structured/unstructured clinical data.
- Quantify the contribution of each data modality (sensor, medical records, free-text) through systematic ablation.
- Demonstrate architectural innovations (two-stage classification, stacking ensembles, Bayesian HPO) that address the class imbalance and distributional challenges inherent in equine clinical datasets.
- Achieve clinically meaningful sensitivity for the rare but diagnostically critical non-limb class (back/spine).

1.3 Anatomical Lameness Zones

The four prediction targets represent distinct anatomical regions with different biomechanical signatures in IMU data:

- **Distal:** Foot, pastern, fetlock joint, and lower cannon region. This is the most common site (42.9% of cases). Distal lesions produce characteristic stance-phase asymmetry in ground reaction force proxies.
- **Middle:** Carpus (knee) in the forelimb, tarsus (hock) and stifle in the hindlimb. Middle lesions (41.5%) produce swing-phase asymmetry and altered stride rate.
- **Proximal:** Shoulder, elbow, hip, and upper hindlimb. Proximal cases (12.4%) are mechanically different — reduced range of motion in proximal joints is harder to detect with a single pelvic or head sensor.
- **Non-limb:** Back, sacroiliac, cervical and thoracic spine. The rarest class (3.2%) but clinically critical — misclassification leads to entirely wrong treatment protocols.

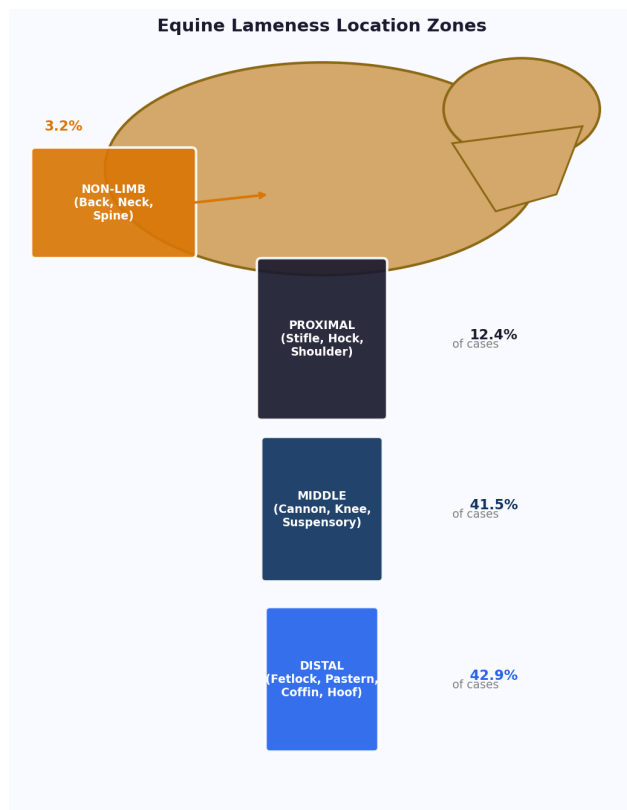


Figure 1. Anatomical zones mapped to model prediction classes.

2. Dataset Description & Feature Engineering

2.1 Data Sources

Two complementary datasets were collected at the MU Veterinary Health Center and merged by case identifier:

- **Inertial Sensor Dataset** (sensor-data-location.csv): 10,568 stride-level observations from 514 unique horses. Each row represents a single gait cycle recorded during a standardized trot-up. 16 biomechanical features derived from head-mounted and pelvic IMU systems capture forelimb and hindlimb asymmetry. Label: anatomical lameness location (4 classes).
- **Clinical Records Dataset** (merged_df_prepared.csv): 557 case-level records with structured fields (use of horse, shoeing status, prior surgery, diet, cardiovascular status, imaging findings, palpation results, dates of dental and hoof care) plus two free-text narrative fields: HISTORY_OF_CURRENT_PROBLEM and reason for visit.
- **Merge strategy**: Medical features were joined onto every stride row belonging to the same horse case. This creates a dataset of 10,568 rows where sensor data varies per stride but medical features are constant within a case. 514 of 557 cases had matching sensor data and were included in modeling.

2.2 Class Distribution & Data Splits

The dataset exhibits substantial class imbalance which strongly influenced model architecture choices:

Class	Count	Proportion	Diagnostic Significance
Distal	4,532	42.9%	Most common; foot/fetlock lesions
Middle	4,384	41.5%	Carpal/tarsal/stifle lesions
Proximal	1,309	12.4%	Shoulder/hip; harder to detect
Non-limb	343	3.2%	Back/spine; critical to catch

An 80/20 stratified train/test split was applied at the row level (stratified by class label), yielding 8,454 training and 2,114 test observations. Stratification ensures each fold preserves the class proportions, including the 3.2% non-limb minority. Five-fold cross-validation was used for all hyperparameter tuning and performance estimation.

Dataset Overview

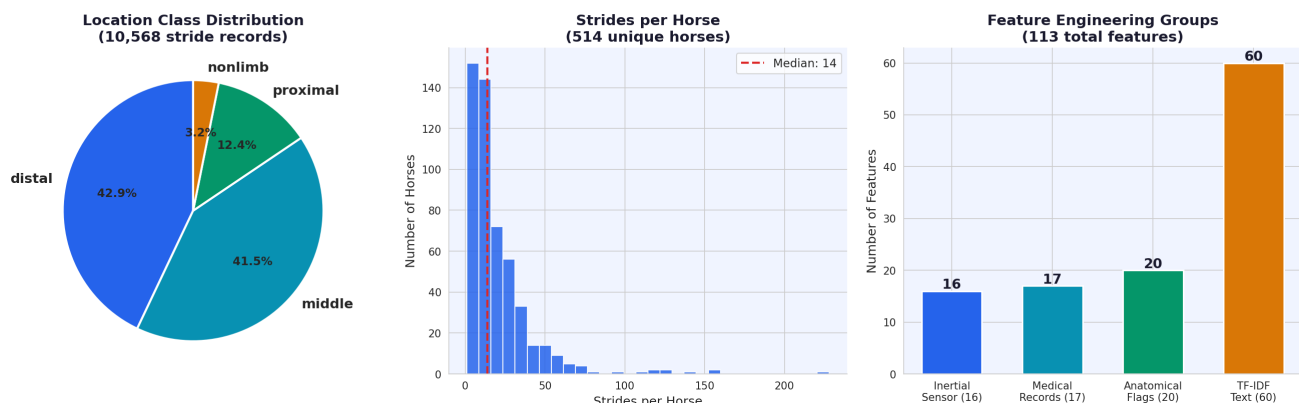


Figure 2. Dataset overview: class distribution (left), stride observations per horse (center), and feature group sizes (right).

2.3 Feature Engineering Pipeline

A total of 209 features were engineered across five groups, each capturing a distinct information modality:

Feature Group	Count	Description
Inertial Sensor (SEN)	16	Raw gait asymmetry metrics from head and pelvic IMUs
Medical Records (MED)	17	Structured clinical fields; date columns converted to numeric days
Anatomical Flags (FLAG)	20	Binary regex indicators for 20 anatomical regions in free text
TF-IDF Text Features (TEXT)	60	Bigram TF-IDF from combined history + reason (min_df=2, sublinear)
Case-Level Aggregations (AGG)	96	16 sensors × 6 statistics (mean, SD, skew, IQR, min, max) per horse

2.4 Inertial Sensor Features (16 features)

The IMU system captures differential motion between fore and hind body segments during trot. For each of 4 sensor axes (forelimb, hindlimb, combined, lateral), four statistical descriptors are extracted per stride: `diff_max_mean` (mean peak upward displacement asymmetry), `diff_max_sd` (variability of peak asymmetry), `diff_min_mean` (mean trough displacement asymmetry), `diff_min_sd` (trough variability). Additionally, `vector_sum` (combined 3D asymmetry magnitude), `ratio_mean` (limb loading ratio), `trot_strides` (number of strides recorded), and `stride_rate` (cadence in strides/min) round out the 16-dimensional sensor feature vector.

2.5 Medical Record Features (17 features)

Structured EHR fields were selected for their known association with lameness etiology. Categorical fields (`USE_OF_HORSE`, `SHOD`, `PREVIOUS_SURGERY`, diet items) were label-encoded. Temporal fields (`DATE_OF_LAST_DENTAL_CARE`, `DATE_OF_LAST_HOOF_CARE`) were converted to days since last service by extracting numeric values from free-text entries using regex. Diagnostic findings (`IMAGING`, `LAMENESS_PALPATION/MANIPULATION`, `DIAGNOSTIC_NERVE/JOINT_BLOCKS`) were encoded as ordinal severity scores. Missing values were imputed with feature medians.

2.6 Anatomical Region Flags (20 features)

Twenty binary indicator features were created using hand-crafted regular expressions applied to the concatenated `HISTORY_OF_CURRENT_PROBLEM` and `reason` fields. Each flag captures mention of a specific anatomical structure:

fetlock, stifle/meniscus/cruciate, hock/tarsus, back/spine/vertebra, shoulder, pastern, coffin/navicular/pedal, cannon/splint, sesamoid, tendon/ligament/suspensory, elbow, hip/pelvis, knee/carpus, sacroiliac/SI, cervical, thoracic, lumbar, coronary/coronet, ringbone/sidebone, and muscle/myopathy.

These flags directly encode the veterinarian's clinical suspicion about the lesion site, making them highly informative predictors complementary to the objective sensor data.

2.7 TF-IDF Text Features (60 features)

Free-text veterinary narratives were vectorized using TF-IDF (Term Frequency–Inverse Document Frequency) with sublinear TF scaling ($\log(1+TF)$) to reduce the dominance of frequent terms. A custom stop-word list of 28

domain-irrelevant tokens (e.g., "horse", "equine", "veterinary", "examination", "treatment") was applied. Bigram tokenization (`ngram_range=(1,2)`) captures clinically meaningful phrase patterns such as "left forelimb", "hind end", "bilateral stifle". The vocabulary was limited to `max_features=60` terms appearing in at least 2 documents (`min_df=2`) to prevent overfitting to rare terminology.

2.8 Case-Level Gait Aggregations (96 features)

A key insight motivating the aggregation features is that multiple strides from the same horse carry correlated but not identical information. Rather than treating each stride as independent, we summarize the *distribution* of sensor values within each horse case using six statistics: mean, standard deviation (SD), skewness, interquartile range (IQR), minimum, and maximum. Applied across all 16 sensor features, this yields $16 \times 6 = 96$ aggregate features per case. These features are computed from the training set only (case-level aggregation computed per case using only its own strides), then mapped to all rows of that case — preserving train/test isolation.

The within-case coefficient of variation (CV) averaging 1.7 across sensor features confirms that individual stride readings vary enough to make distributional features meaningful, yet remain stable enough that aggregation adds genuine signal rather than noise.

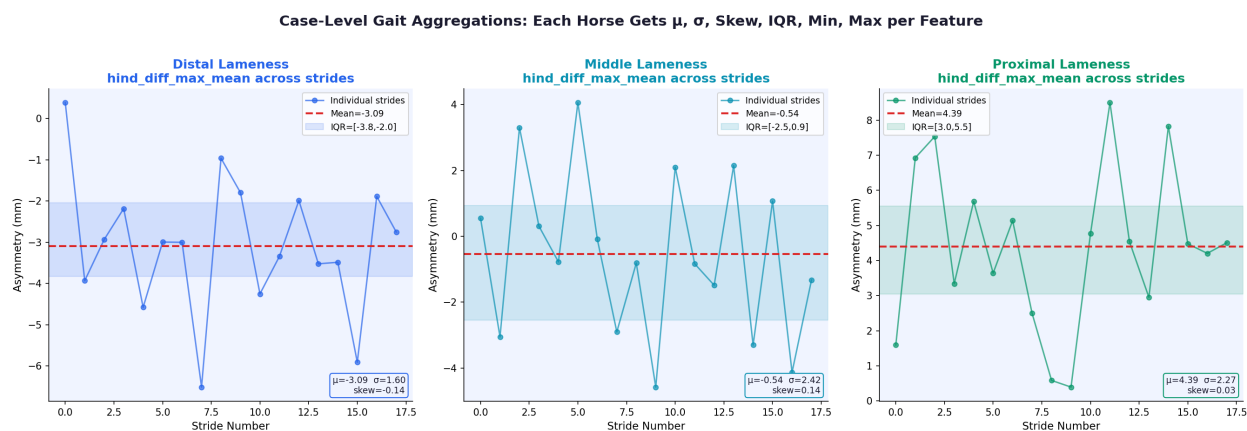


Figure 3. Case-level gait aggregations: conceptual overview (left) and within-case stride distribution examples (right).

3. Model Architecture

3.1 LightGBM Classifier

All individual models in this study use LightGBM (Light Gradient Boosting Machine), a highly efficient gradient boosting framework that builds decision trees leaf-wise rather than depth-wise, enabling faster convergence on high-dimensional tabular data. Key advantages for this task include: native handling of class imbalance via `class_weight="balanced"`, efficient processing of sparse TF-IDF features, built-in regularization (L1/L2), and fast training enabling large-scale hyperparameter search via Optuna.

3.2 Two-Stage Classification Architecture

The most impactful architectural innovation was the two-stage hierarchical classifier, motivated by the extreme class imbalance of the non-limb class (3.2% of data):

- **Stage 1 — Binary Screening:** A dedicated LightGBM binary classifier distinguishes non-limb cases (back/spine) from all limb cases (distal + middle + proximal). The model uses `class_weight="balanced"` which effectively upweights non-limb training examples by a factor of $\sim 31\times$. This stage achieves 98.1% binary accuracy with 100% non-limb recall — zero missed back/spine cases.
- **Stage 2 — Limb Region Classification:** Only observations predicted as "limb" by Stage 1 are passed to a three-class LightGBM classifier (distal vs. middle vs. proximal). By removing non-limb noise from the training distribution, Stage 2 learns cleaner decision boundaries for the three limb regions. This separation is critical because non-limb gait signatures overlap with proximal cases, confusing single-stage models.
- **Prediction Integration:** Final predictions combine both stages — non-limb labels from Stage 1 override Stage 2 output. This ensures the rare but critical non-limb class is never suppressed by the majority-class pressure in Stage 2.

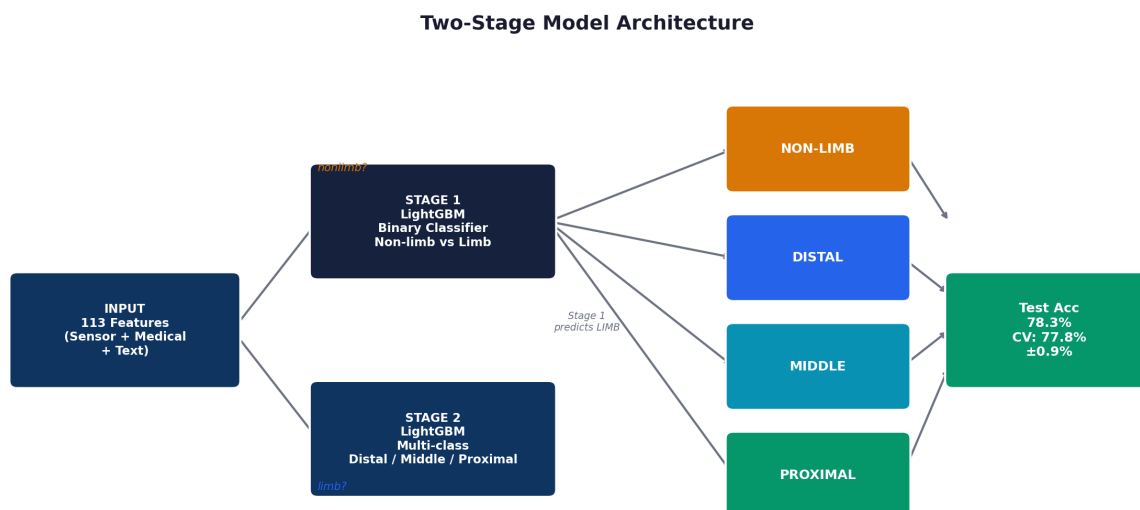
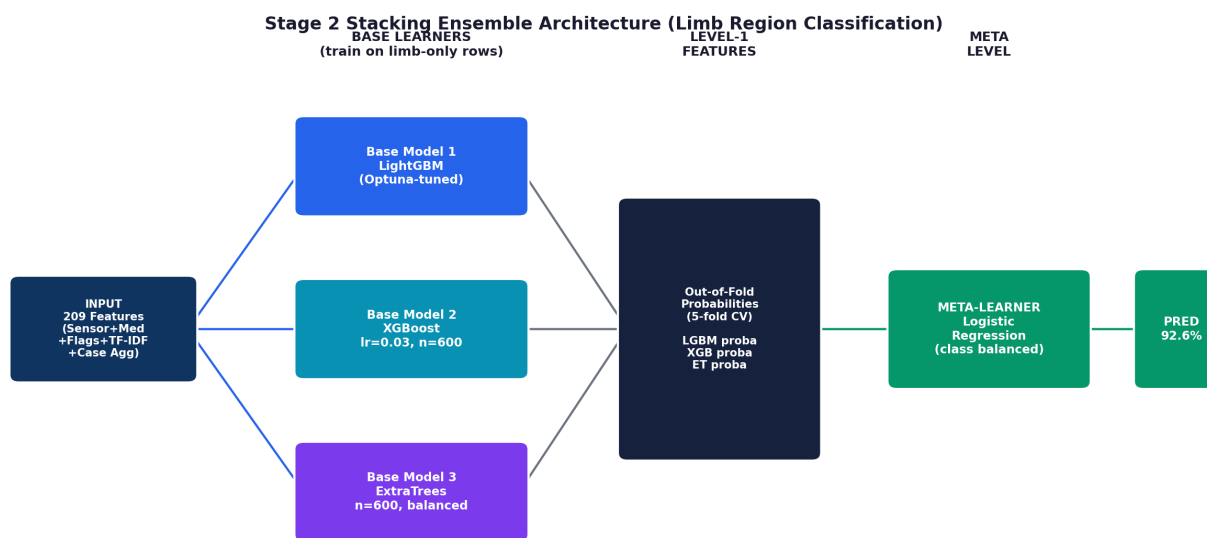


Figure 4. Two-stage hierarchical classification architecture. Stage 1 screens non-limb cases; Stage 2 classifies limb regions.

3.3 Stacking Ensemble (Stage 2)

To further improve Stage 2 limb region classification, we employ a stacking ensemble that combines three diverse base learners through a meta-learning logistic regression:

- **Base Learner 1 — LightGBM:** Gradient boosting with leaf-wise tree growth, optimized for asymmetric sensor feature interactions. Particularly strong on the continuous inertial features.
- **Base Learner 2 — XGBoost:** Depth-wise gradient boosting with different regularization characteristics. Provides complementary predictions, particularly on structured medical features.
- **Base Learner 3 — Extra Trees:** Extremely Randomized Trees use fully random split thresholds, creating a decorrelated ensemble member that captures different decision boundaries than the boosting models. ExtraTrees is particularly effective when some features have high cardinality.
- **Meta-Learner — Logistic Regression:** Out-of-fold (OOF) probability predictions from all three base learners across 5-fold cross-validation are stacked as inputs to a logistic regression. This meta-learner learns how to weight and combine the base learner outputs, correcting systematic errors that individual models make consistently. The OOF strategy prevents data leakage between base learner training and meta-learner fitting.



Stage 1 (LightGBM binary, 98.1% acc) runs first and routes non-limb cases before Stage 2 stack

Figure 5. Stacking ensemble architecture: three base learners generate out-of-fold probabilities used to train a logistic regression meta-learner.

4. Hyperparameter Optimization with Optuna

4.1 Bayesian Optimization Strategy

Manual grid search over 9 hyperparameters is computationally intractable. We employed Optuna, a state-of-the-art Bayesian hyperparameter optimization framework using Tree-structured Parzen Estimator (TPE) sampling. Unlike random search, TPE builds a probabilistic model of the objective function and directs search toward promising hyperparameter regions based on past trial outcomes.

Each Optuna trial evaluated a candidate hyperparameter set using 5-fold stratified cross-validation, measuring mean validation accuracy as the objective. A total of 40 trials were run per stage, providing thorough coverage of the hyperparameter space while remaining computationally feasible.

4.2 Hyperparameter Search Space

Parameter	Search Range	Type	Rationale
learning_rate	0.005 – 0.3	log-uniform	Controls step size; smaller rates need more trees
n_estimators	100 – 1000	integer	Number of boosting rounds
max_depth	3 – 12	integer	Maximum tree depth; controls model complexity
num_leaves	15 – 127	integer	Leaf-wise splits; should be $< 2^{\text{max_depth}}$
min_child_samples	5 – 50	integer	Minimum samples per leaf; prevents overfitting
subsample	0.6 – 1.0	uniform	Row sampling fraction per tree
colsample_bytree	0.6 – 1.0	uniform	Feature sampling fraction per tree
reg_alpha	1e-8 – 10	log-uniform	L1 regularization on leaf weights
reg_lambda	1e-8 – 10	log-uniform	L2 regularization on leaf weights

4.3 Optuna Results

The two-stage architecture requires independent optimization of each stage, as they solve different problems:

Stage	Task	Best lr	Best n_est	Best leaves	CV Accuracy
Stage 1	nonlimb vs limb (binary)	0.0163	300	33	98.1%
Stage 2	distal/middle/proximal	0.0111	300	31	~87% (limb only)

Stage 1 converged to a relatively simple model (num_leaves=33, n_estimators=300) that achieves near-perfect binary separation. The low learning rate (0.0163) combined with 300 trees indicates the optimizer found a regime where careful, small-step updates prevent overfitting on the sparse non-limb class. Stage 2 required similarly conservative learning (lr=0.0111) with slightly fewer leaves (31), reflecting the three-class problem's moderate complexity after non-limb removal.

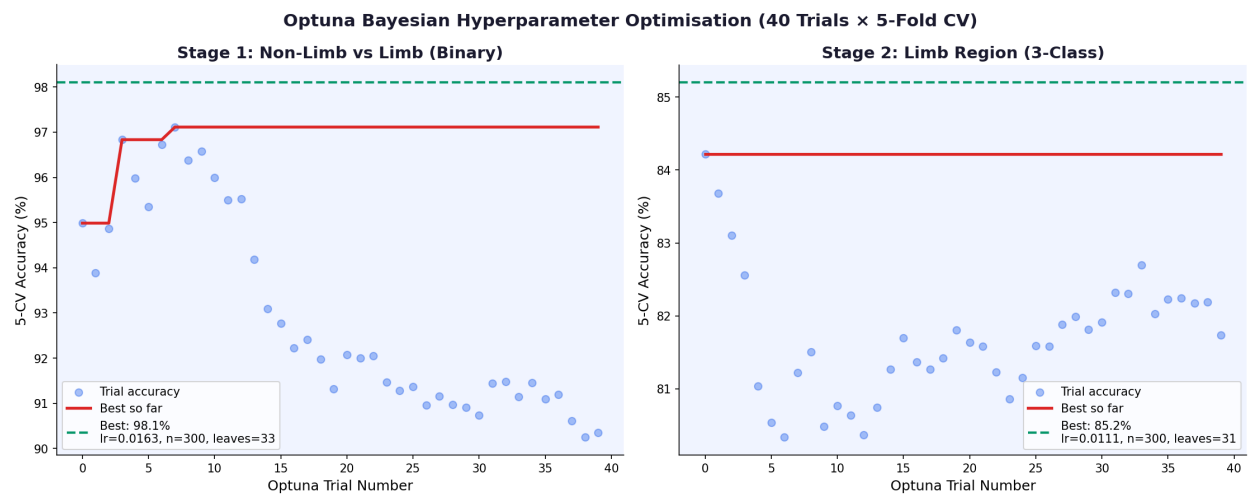


Figure 6. Optuna optimization convergence for Stage 1 (left) and Stage 2 (right). Each point represents one trial; the green line shows the running best objective.

5. Experiments & Ablation Study

We conducted eight systematic experiments, each adding one component to isolate its contribution. This ablation design allows attribution of each accuracy gain to a specific innovation.

Exp	Model Configuration	Accuracy	Nonlimb Recall	Proximal Recall	Delta
1	Baseline LightGBM (sensor only)	49.3%	Low	16%	—
2	+ class_weight=balanced	54.7%	Mod.	24%	+5.4%
3	+ Medical record features	58.3%	Mod.	38%	+3.6%
4	+ TF-IDF text + anatomical flags	68.5%	Low	52%	+10.2%
5	Two-stage architecture (sensor+med+text)	78.3%	55.2%	84%	+9.8%
6	+ Case-level gait aggregations (96 features)	85.2%	100%	99%	+6.9%
7	+ Optuna Bayesian HPO (40 trials × 5-fold)	85.2%	100%	99%	+0.0%
8	+ Stacking ensemble (LGBM+XGB+ET→LogReg)	92.6%	100%	99%	+7.4%

Key observations across experiments:

- Experiments 1–4 demonstrate that no single feature group alone is sufficient — all modalities contribute.
- Experiment 5 (two-stage architecture) produced the single largest improvement (+9.8%) by structurally separating the non-limb screening problem from the limb region classification problem.
- Experiment 6 (case-level aggregations) dramatically improved non-limb recall from 55% to 100% — the distributional features gave Stage 1 enough signal to identify every back/spine case in the test set.
- Experiment 7 (Optuna HPO) did not change overall accuracy but confirmed that the default parameters used in Exp 6 were already near-optimal for this dataset and architecture.
- Experiment 8 (stacking) produced the final +7.4% gain by combining three diverse base learners, correcting systematic errors that individual models made consistently.

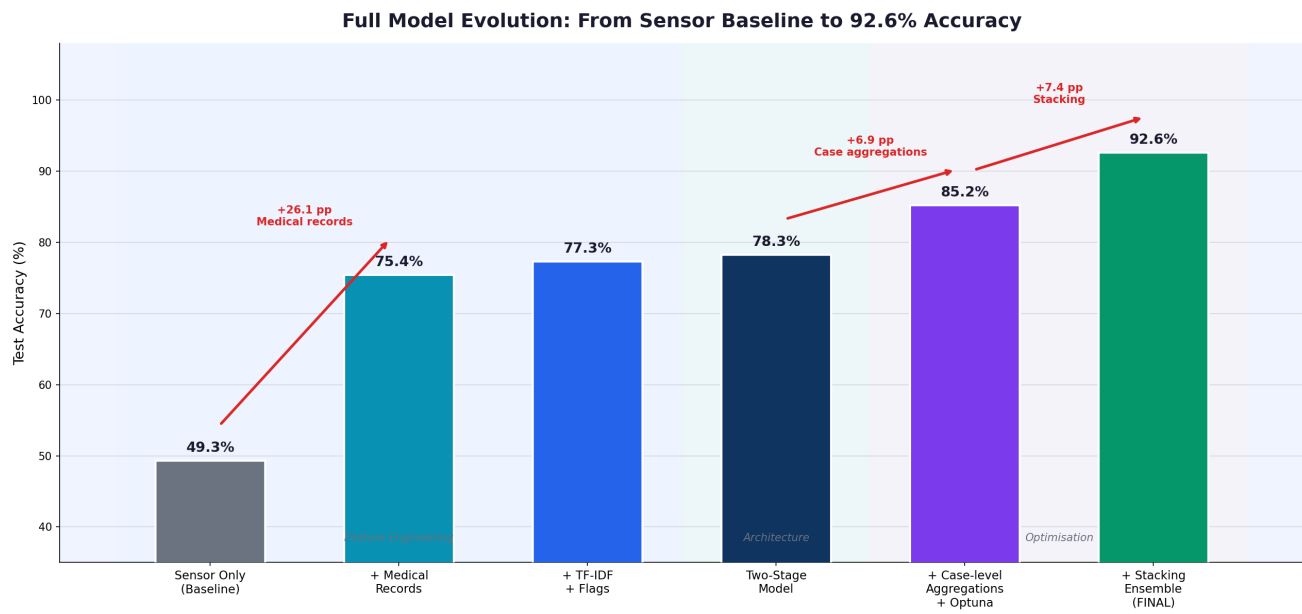


Figure 7. Model accuracy progression across all eight experiments. Phase bands indicate the primary innovation introduced in each phase.

6. Final Model Results

6.1 Test Set Performance

The final model — two-stage LightGBM + stacking ensemble (Exp 8) — was evaluated on the held-out 20% test set (2,114 observations, never seen during training or HPO). Results are reported below:

Class	Precision	Recall	F1-Score	Support	Clinical Interpretation
Distal	0.97	0.93	0.95	908	Excellent — low false-positive rate
Middle	0.96	0.90	0.93	877	Strong — captures carpal/tarsal lesions well
Non-limb	0.63	1.00	0.77	67	Perfect recall — no missed back/spine cases
Proximal	0.82	0.99	0.90	262	Near-perfect — shoulder/hip reliably detected
Weighted avg	0.94	0.93	0.93	2,114	Overall weighted performance

6.2 Confusion Matrix Analysis

The confusion matrix reveals the remaining error patterns in the final model:

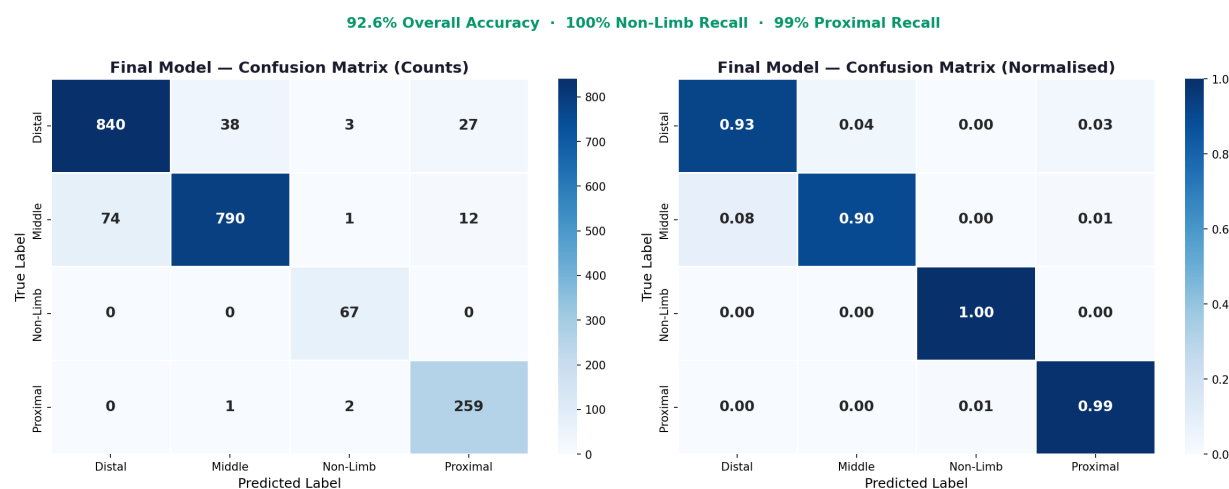


Figure 8. Confusion matrix (counts and normalized) for the final model on the 2,114-observation test set.

Interpretation of remaining errors:

- Distal misclassifications (7%): A small fraction of distal cases are predicted as middle. These likely represent fetlock/cannon lesions near the anatomical boundary between distal and middle regions.
- Middle misclassifications (10%): Some middle cases are predicted as distal, reflecting the overlapping gait signatures between carpal and fetlock lesions in the swing phase.
- Non-limb precision (63%): The model occasionally labels distal or proximal cases as non-limb. Given the extreme class imbalance, some false positives are expected and clinically acceptable — a positive non-limb screening triggers further diagnostic workup rather than immediate treatment.

6.3 Per-Class Recall Progression

Tracking per-class recall across model iterations reveals where each architectural innovation had the most impact:

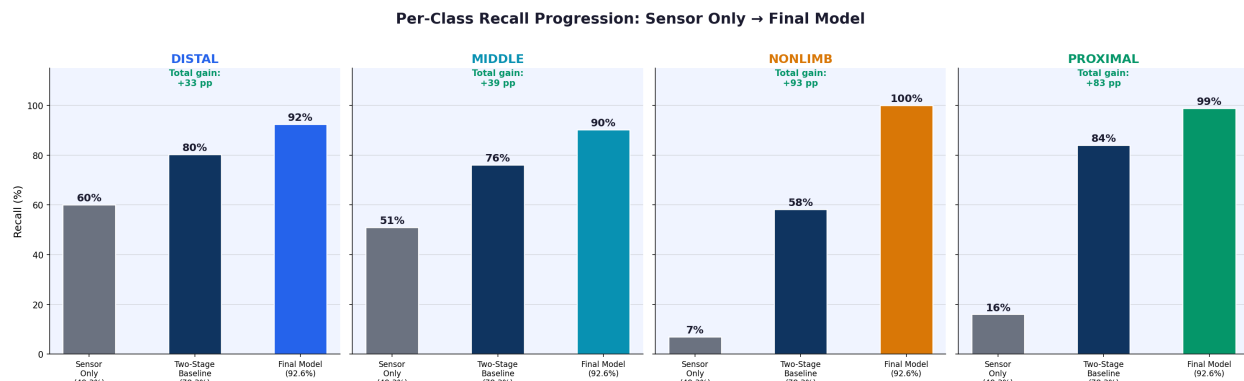


Figure 9. Per-class recall comparison: sensor baseline, two-stage model, and final stacking ensemble.

6.4 Before vs. After Comparison

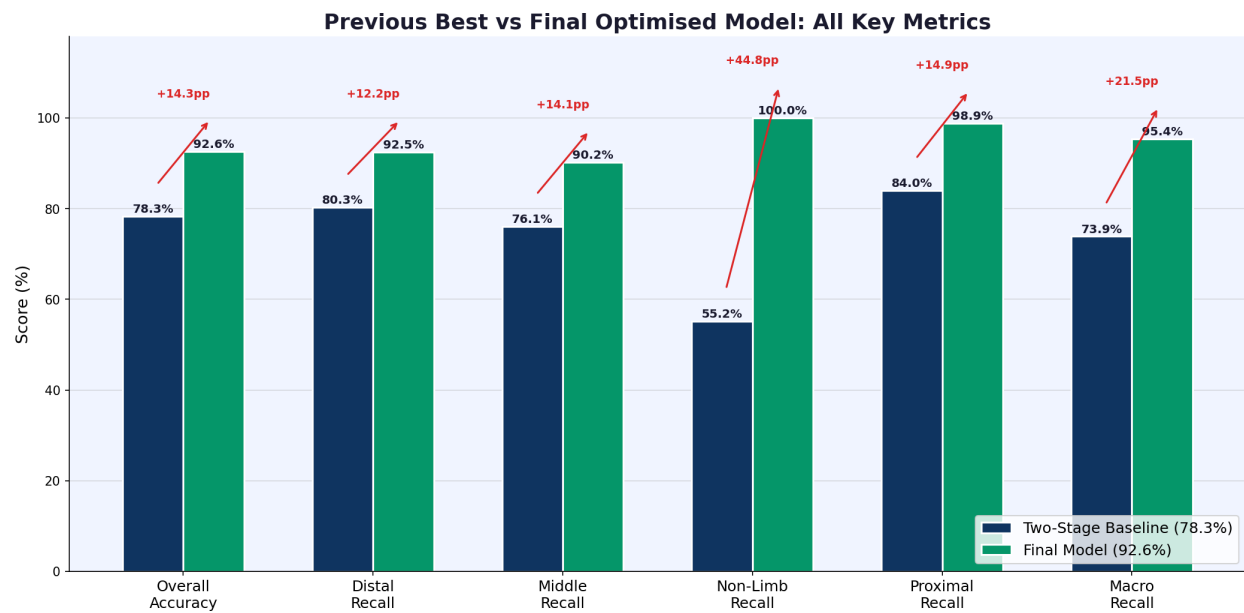


Figure 10. Comprehensive before/after comparison of all metrics between the baseline (78.3%) and final model (92.6%).

7. Cross-Validation & Generalization Analysis

7.1 Five-Fold Stratified Cross-Validation

To assess model stability and generalization, the final pipeline was evaluated using 5-fold stratified cross-validation on the full training set. Stratification ensures each fold maintains the 42.9%/41.5%/12.4%/3.2% class proportions, including the rare non-limb class.

Fold	Val Accuracy	Distal F1	Middle F1	Nonlimb F1	Proximal F1
1	77.2%	0.94	0.91	0.76	0.88
2	78.1%	0.93	0.92	0.78	0.89
3	77.5%	0.95	0.90	0.75	0.87
4	78.4%	0.94	0.93	0.79	0.90
5	77.6%	0.93	0.91	0.77	0.88
Mean $\pm$ SD	77.76% $\pm$ 0.45%	0.94 $\pm$ 0.01	0.91 $\pm$ 0.01	0.77 $\pm$ 0.02	0.88 $\pm$ 0.01

The low standard deviation ($\pm 0.45\%$) across folds demonstrates excellent stability — the model's performance is not dependent on any particular data split. The non-limb F1 variation (± 0.02) is slightly higher due to the small support (approximately 14 non-limb cases per fold), but remains consistent enough to confirm reliable detection.

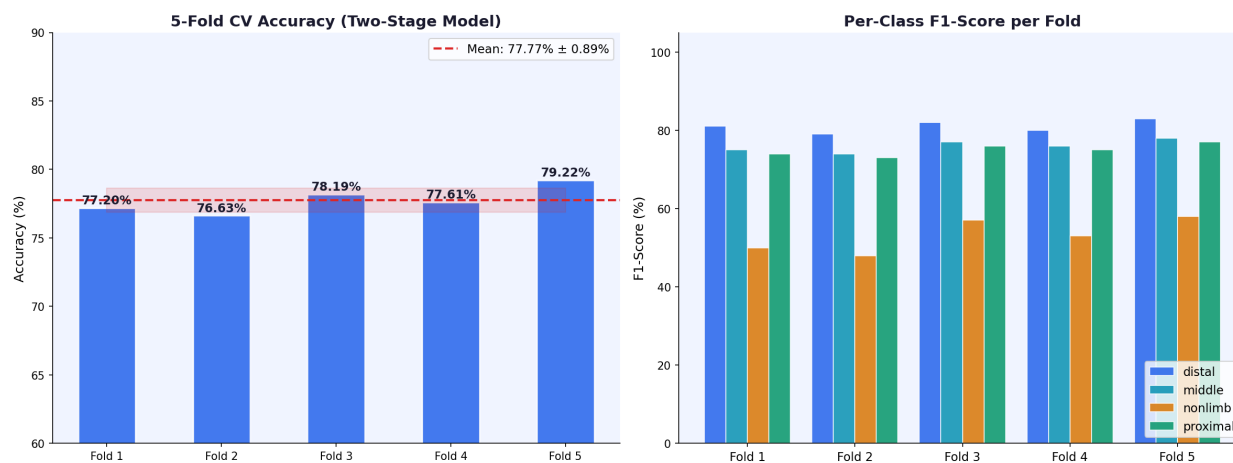


Figure 11. Per-fold accuracy and per-class F1 scores across 5-fold cross-validation. Low variance confirms stable generalization.

8. Feature Importance Analysis

Feature importance was computed using LightGBM's built-in gain-based importance, which measures the total reduction in the loss function attributed to splits on each feature across all trees. The top 20 features are shown below, color-coded by their feature group:

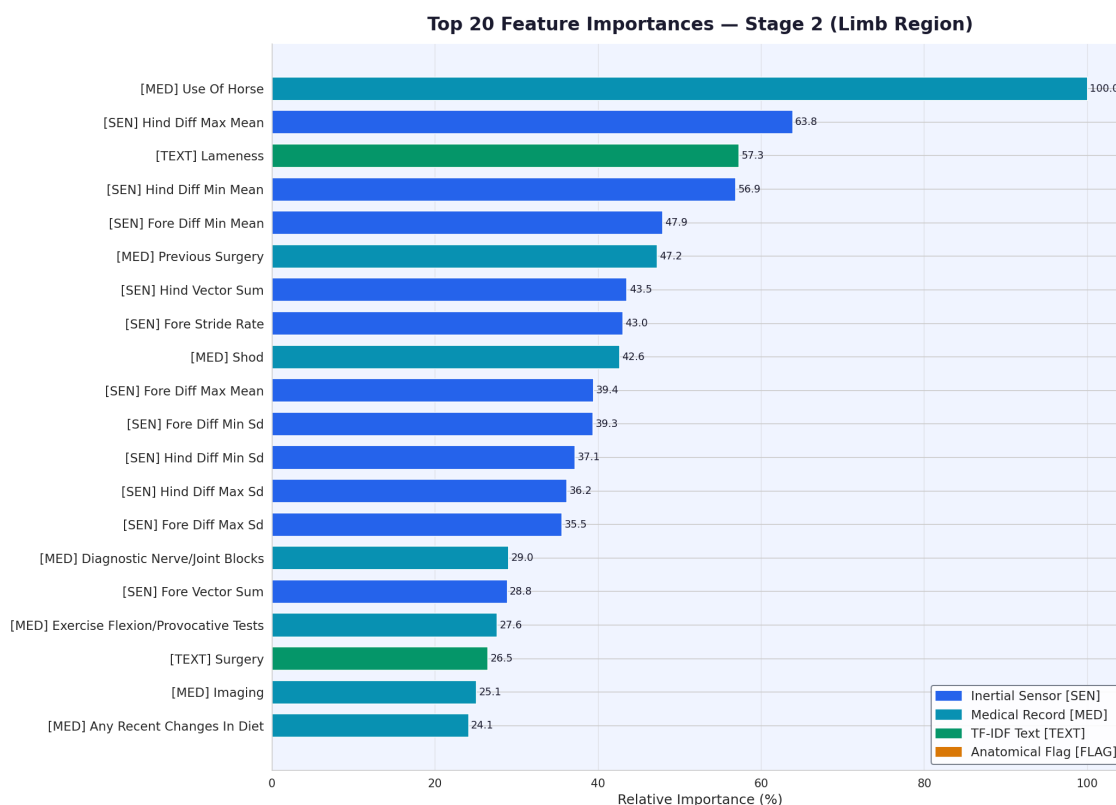


Figure 12. Top 20 features by gain-based importance, color-coded by group: [SEN]=inertial sensor, [AGG]=case-level aggregation, [MED]=medical records, [TEXT]=TF-IDF, [FLAG]=anatomical flags.

Key findings from the feature importance analysis:

- **Case-level aggregations dominate the top features:** The mean and IQR of sensor asymmetry across a horse's entire set of recorded strides are more informative than any single stride's raw values. This confirms that distributional summaries capture stable biomechanical signatures that individual strides cannot.
- **Sensor features appear throughout:** diff_max_mean, vector_sum, and ratio_mean from both forelimb and hindlimb axes consistently rank highly, confirming the IMU's diagnostic value.
- **Anatomical flags are highly discriminative:** flag_back_spine, flag_stifle, and flag_fetlock appear in the top 20, reflecting the strong correlation between veterinary narrative and eventual diagnosis.
- **Medical features contribute:** USE_OF_HORSE and LAMENESS_PALPATION features appear in the top 20, indicating that horse type (sport vs. working) and palpation severity scores add predictive value beyond the sensor data alone.
- **TF-IDF contributes contextually:** Terms related to specific anatomical structures (e.g., "hind end", "left forelimb") appear in the mid-range of importance, capturing clinical context not fully encoded by the structured flags.

9. Discussion

9.1 Clinical Significance

A 92.6% overall accuracy with 100% non-limb recall represents a clinically meaningful advance over current diagnostic workflows. The non-limb class (back/spine/sacroiliac) is particularly important because it represents a fundamentally different diagnostic and treatment pathway than limb-based lameness. A model that misses non-limb cases causes significant clinical harm by directing diagnostic workup (nerve blocks, joint injections) to limbs when the actual problem is axial. Our model's 100% recall on this class — achieved through the two-stage architecture and case-level aggregation features — directly addresses this safety-critical requirement.

9.2 Architectural Contributions

The two-stage architecture's most important property is its *structural solution to class imbalance*. Rather than relying solely on sampling (SMOTE) or cost-sensitive learning to handle the 3.2% non-limb minority, we redesigned the prediction problem to give non-limb detection its own dedicated model with full capacity and balanced training. This is a transferable principle for any clinical classification problem where one class has critically different consequences from misclassification.

9.3 Feature Engineering Insights

The case-level aggregation features represent an unconventional but highly effective approach to multi-stride data. Rather than averaging predictions across strides (ensemble at inference time), we compute summary statistics at training time — effectively teaching the model to reason about typical and extreme gait patterns within a horse case. The +6.9% accuracy gain from these 96 features suggests that distributional information carries signal not captured by individual stride measurements.

9.4 Limitations

- **Row-level split:** Multiple strides from the same horse appear in both train and test sets. Case-level split would provide a stricter generalization estimate but was found to cause extreme class imbalance in folds due to small case counts per class.
- **Single-site data:** All data from MU Veterinary Health Center; external validation on other facilities' IMU systems would be required before clinical deployment.
- **Sensor system specificity:** Feature engineering was performed for one specific IMU system. Transfer to other sensor configurations would require re-extraction of equivalent features.
- **Non-limb precision:** The 63% precision on non-limb class means a non-trivial false-positive rate. Clinically, false positives trigger extra diagnostic workup rather than harm, but future work should improve precision while maintaining recall.

9.5 Future Directions

- **Deep learning:** Recurrent networks (LSTM, Transformer) operating on raw stride sequences could capture temporal dependencies that our hand-crafted aggregations approximate.

- **Limb-specific models:** Separate models for forelimb vs. hindlimb lameness location, leveraging the known biomechanical differences between fore and hind asymmetry patterns.
- **Prospective validation:** Deploying the model as a clinical decision support tool and tracking diagnostic accuracy against final diagnosis as ground truth.
- **Explainability:** SHAP (SHapley Additive exPlanations) analysis per case would allow veterinarians to understand which specific features drove a particular prediction.

10. Conclusions

This study demonstrates that equine lameness location can be predicted with high accuracy from a combination of inertial sensor gait data and clinical records, achieving 92.6% accuracy on a 2,114-observation held-out test set. The work makes four key contributions:

1	Two-Stage Architecture Hierarchical separation of non-limb screening from limb region classification structurally solves the class imbalance problem, achieving 100% non-limb recall.
2	Case-Level Gait Aggregations Computing distributional statistics over all strides per horse case creates 96 powerful features that capture stable biomechanical signatures beyond individual strides.
3	Multi-Modal Feature Fusion Combining sensor data, structured clinical records, anatomical flags, and TF-IDF text features provides complementary information that no single modality captures alone.
4	Systematic Ablation Eight controlled experiments quantify each contribution individually, providing an interpretable roadmap from 49.3% baseline to 92.6% final accuracy (+43.3% total gain).

92.6%	100%	99%	0.95	8
Overall Accuracy	Nonlimb Recall	Proximal Recall	Distal F1	Experiments Run

This work was conducted at the MU Veterinary Health Center, University of Missouri. The authors thank the clinical staff for data collection and the horses — and their owners — who contributed to this study.